

## 1.7 Linear Independence

### Definition

An indexed set of vectors  $\{\vec{v}_1, \dots, \vec{v}_p\}$  in  $\mathbb{R}^n$  is said to be linearly independent if the vector equation

$$x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_p \vec{v}_p = \vec{0}$$

has only the trivial solution. The set  $\{\vec{v}_1, \dots, \vec{v}_p\}$  is said to be linearly dependent if there exists weight  $c_1, \dots, c_p$ , not all zeros, such that

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_p \vec{v}_p = \vec{0} \quad (2)$$

p. 84

Equation (2) is called a linear dependence relation among  $\vec{v}_1, \dots, \vec{v}_p$  when the weights are not all zero.

The columns of the matrix  $A$  are linearly independent if and only if the equation  $A\vec{x} = \vec{0}$  has only the trivial solution.

### Theorem 7 - Characterization of Linearly Dependent Sets

An indexed set  $S = \{\vec{v}_1, \dots, \vec{v}_p\}$  of two or more vectors is linearly dependent if and only if at least one of the vectors in  $S$  is a linear combination of the others. In fact, if  $S$  is linearly dependent and  $\vec{v}_1 \neq \vec{0}$ , then some  $\vec{v}_j$  (with  $j > 1$ ) is a linear combination of the preceding vectors,  $\vec{v}_1, \dots, \vec{v}_{j-1}$ .

p. 86

### Theorem 8

If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set  $\{\vec{v}_1, \dots, \vec{v}_p\}$  in  $\mathbb{R}^n$  is linearly dependent if  $p > n$ .

p. 87

### Theorem 9

If a set  $S = \{\vec{v}_1, \dots, \vec{v}_p\}$  in  $\mathbb{R}^n$  contains the zero vector, then the set is linearly dependent.

p. 87